

This box is for the examiner only:

Question:	1	2	3	4	5	Total
Points:	12	27	17	24	10	90
Score:						

Analytical Skills for Business (01/2026)

1. Version Control Systems: Git and GitHub

- (a) (6 points) Explain the key differences between Git and GitHub. What is the primary purpose of each?
- (b) (6 points) Describe three core concepts in Git—repository, commit, and branch—and explain their importance in collaborative analytics projects.

2. Programming Languages and Development Environments

- (a) (6 points) Compare Python and SQL in terms of their primary use cases in business analytics.
- (b) (9 points) Explain three key elements of programming languages (choose from syntax, libraries, variables, functions, objects, conditions, loops) and describe why each is important for data analysis.
- (c) (6 points) Define what containers like Docker do in software development and data analytics. Explain the main benefits they provide.
- (d) (6 points) Explain the concept of an IDE and name two popular IDEs used for data analytics.

3. No-Code and Low-Code Tools

- (a) (9 points) Name and describe three no-code or low-code tools for data analytics covered in the course (choose from n8n, Snowflake, QGIS, Tableau, Power BI, KNIME).
- (b) (8 points) Explain the advantages of using no-code/low-code tools in business analytics and discuss situations where these tools might be preferred over traditional programming methods.

4. Statistics and Analytics

- (a) (9 points) Explain three measures of central tendency (mean, median, mode) and when each is most appropriate to use.
- (b) (9 points) Distinguish between univariate, bivariate, and multivariate data analysis. Provide one business example for each.
- (c) (6 points) Describe three common data quality issues (from: missing values, inconsistent data, duplicate records, outliers) and appropriate strategies for handling each.

5. Practical R Programming

- (a) (3 points) Write R code to calculate and store the following result in a variable named `result`:

$$3 + \sqrt{9} - 2^2 \cdot \frac{4}{3}$$

- (b) (2 points) Write the R code to get the path of your current working directory.
- (c) (2 points) Write the R code to access the help documentation for the `mean()` function.
- (d) (3 points) Create a data frame in R containing the following sales data:

Region	Sales
North	45000
South	38000
East	52000

Store it in a variable named `sales_data`.